PROOF OF CONCEPT

Vitsika

Malagasy ant identification from a single photograph

vitsika is Malagasy for ant. Genus-level, 27 genera, built on public data and a frozen public model.

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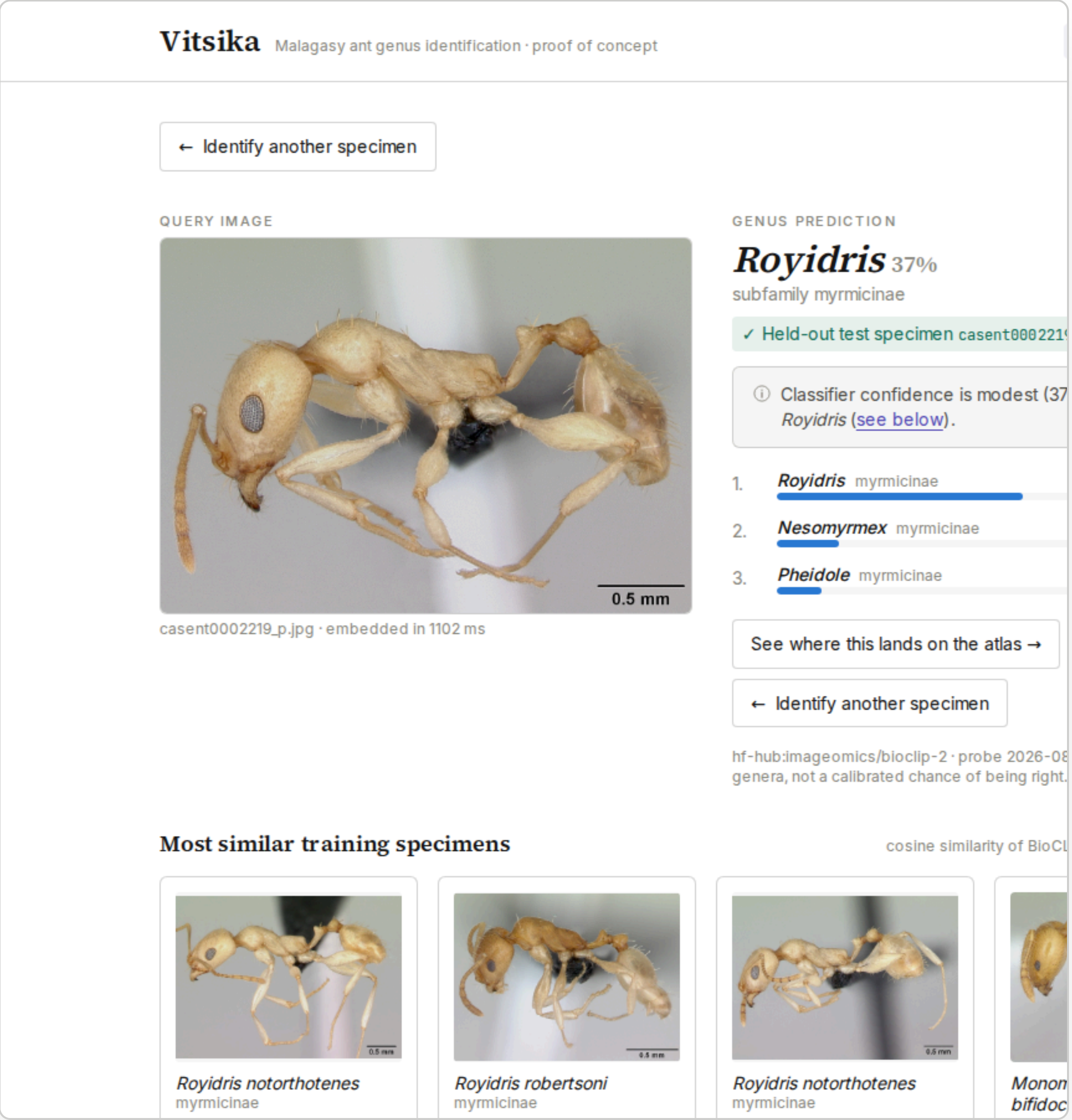


Identification is the bottleneck, and it happens far from the field

- AntWeb holds **141,430** Malagasy ant records, **6,995** of them with images, across **60** imaged genera. Naming even the genus takes a specialist and a microscope.

- The specimens, the imaging and the determinations are curated and published by the California Academy of Sciences (San Francisco); the expertise sits outside Madagascar, and a field team waits for it.

- A photo-to-genus first pass that runs on a CPU box in Antananarivo in **~0.8 s** per image lets a local team triage, then send only the hard cases to a specialist.

One AntWeb profile shot per genus in scope (27). Sources: reports/01_explore.txt, README.md.

From GBIF to 1,236 clean images in 27 genera

141,430
AntWeb records for Madagascar on GBIF (doi:10.15468/wqmjjt); 6,995 with an image

→

4,354
specimens, 39 genera: workers, profile view, one image each, ≥ 10 per genus

→

1,297
images obtained (29.8 %): 1,160 Wikimedia Commons + 137 GBIF cache

→

1,236
images, **27 genera**, 8 subfamilies; 986 train / 250 held-out test

Finding 1: GBIF's image cache is empty for AntWeb

4,215 of 4,354 cache requests returned 404 (96.8 %), uniformly across years and record age. antweb.org blocks scripts (Cloudflare) and datacenter IPs (403), including GBIF's own fetcher. The working source is the 2008-2014 bulk upload to Wikimedia Commons (33,081 files), matched by specimen code.

Finding 2: caste is not in GBIF's interpreted fields

GBIF maps `sex` to Male / Female / Other, so every worker arrives as null: 4,540 of 5,857 specimens. The caste has to be read from the verbatim record (`dwc:sex`): 4,539 workers, 673 queens, 640 males, and only workers are kept.

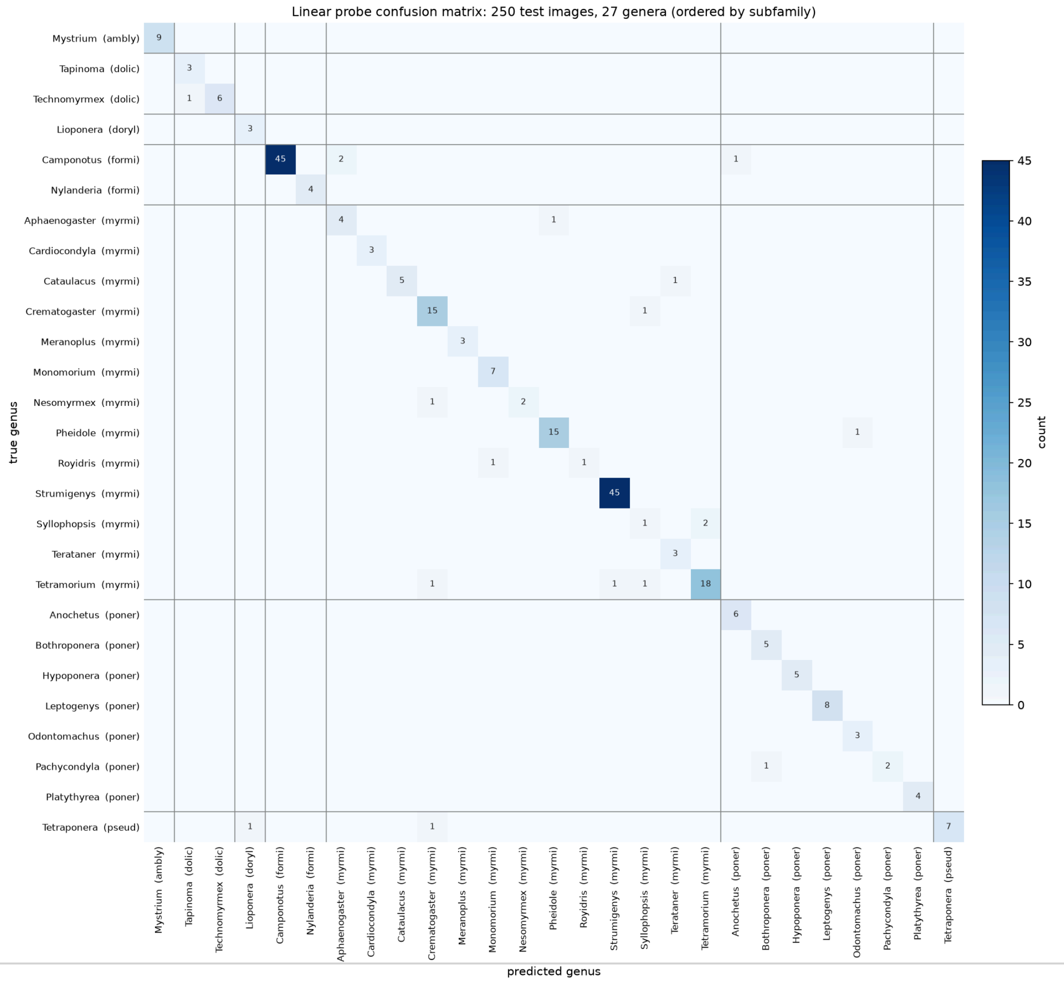
RESULTS

Linear probe on frozen BioCLIP 2 embeddings: 92.8 % top-1 on 250 held-out specimens

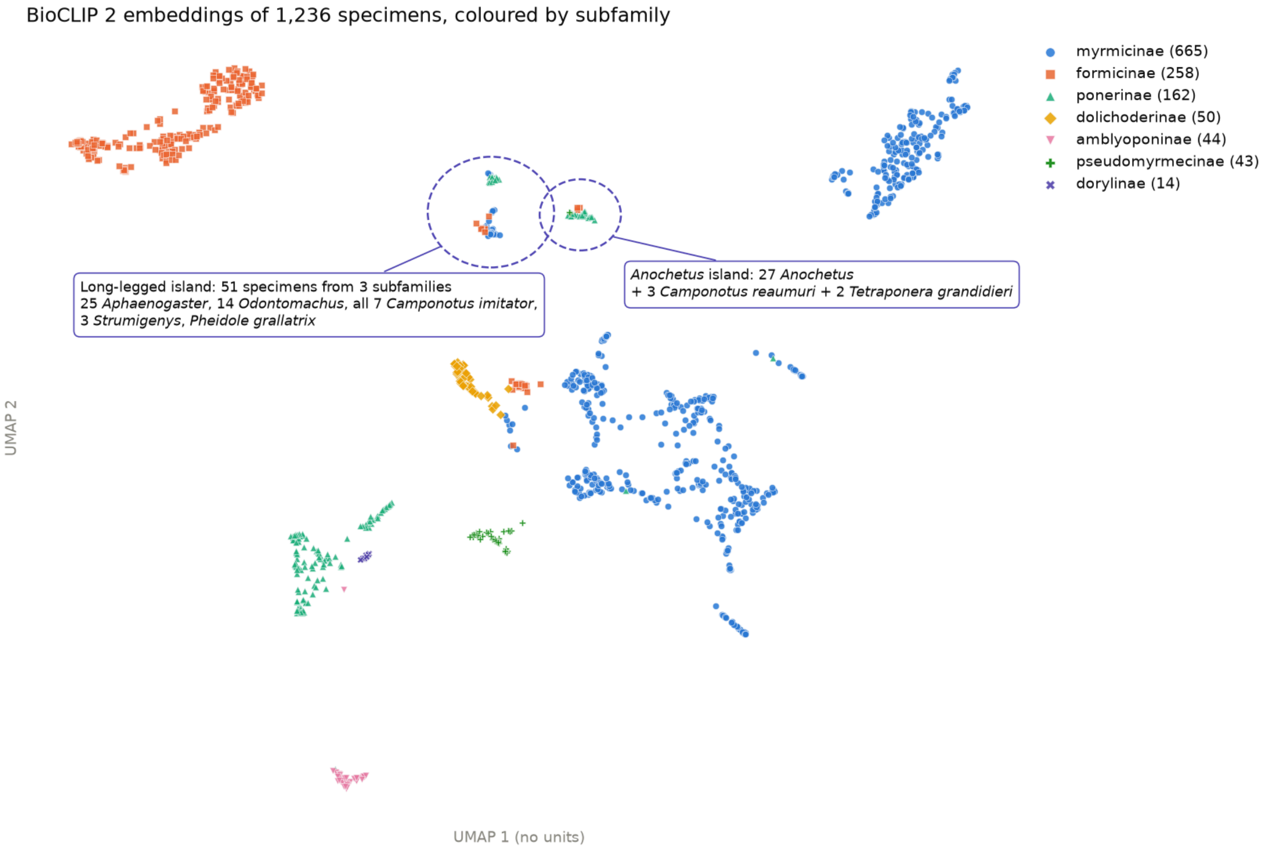
METHOD	TOP-1	TOP-3	MACRO-F1
Zero-shot, “a photo of {genus}, a genus of ant”	74.4 %	94.0 %	0.590
Linear probe (deployed)	**92.8 %**	**98.4 %**	**0.885**
Nearest neighbour, cosine	91.6 %	93.2 %	0.834

986 train / 250 test, 27 genera, split by genus (seed 42). Bare genus name as the zero-shot prompt: 32.8 % top-1. Embeddings: `imageomics/bioclip-2` ViT-L/14, 768-d, no fine-tuning.

> 18 errors: 12 genuine look-alikes or species unseen in training (*Syllophopsis* vs *Tetramorium*, *Royidris* vs *Monomorium*), 3 poor photos, 2 odd angles, 1 unexplained. Almost every confusion stays inside a subfamily block.



The embedding space recapitulates taxonomy, with body-plan islands



Subfamilies form clean clusters; where the model "errs", it groups *Camponotus imitator* with long-legged *Aphaenogaster* and *Odontomachus*, a resemblance a taxonomist would recognise, not noise.

NEXT STEPS

Three things this codebase already supports

1

Complete the image set

70.2 % of the manifest's images are missing. `05_download_antweb.py` + `dataset_full.csv` are ready for a residential IP or an AntWeb bulk export; re-thresholding brings *Vitsika*, *Tanipone*, *Carebara* and 9 more genera into scope (39 in total).

2

Honest generalisation and an open set

Group the split by species (one flag in `03_build_dataset.py`), calibrate the probe's probabilities on the held-out set, and turn the nearest-neighbour similarity the API already returns into a "none of the 27" reject.

3

Multi-view, multi-caste embeddings

The harvest already holds 6,634 head and 5,127 dorsal media rows plus 673 queens and 640 males; `config.yaml` (`views_priority`, `castes`) and `06_embed.py` make view-wise embedding and late fusion a configuration change.

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